

UMER SIDDIQUI

Backend & AI Systems Engineer | System Design

um3rsiddiqui99@gmail.com • +92-341-8094081 • Karachi, Pakistan • [linkedin.com/in/um3r-siddiqui](https://www.linkedin.com/in/um3r-siddiqui)

github.com/UmerSiddiqui • umersiddiqui.github.io/Portfolio

PROFESSIONAL SUMMARY

Backend and AI Systems Engineer with production experience building scalable, low-latency LLM-powered applications — agentic workflows, RAG pipelines, and enterprise voice AI — from architecture through cloud deployment. Strong grounding in core system design (API architecture, database selection, caching, load balancing, authentication/security) combined with applied AI infrastructure experience (model serving, orchestration, LLM Ops, cost and latency trade-offs). Comfortable owning a system end-to-end: designing the architecture, writing the backend, and reasoning about how it behaves at scale. Strong Data Structures & Algorithms fundamentals, applied through consistent problem-solving practice.

Core Stack: Python · FastAPI · Node.js/NestJS · TypeScript · PostgreSQL · pgvector · Redis · OAuth 2.0/JWT · Docker · AWS · GCP · CI/CD · LangGraph · RAG · System Design · DSA

PROFESSIONAL EXPERIENCE

Senior AI Engineer | Saynt AI 06/2025 – Present · Remote

- Owned end-to-end backend development for the voice agent dashboard and a dental practice management (PMS) platform — REST APIs, business logic, and authentication built in NestJS/TypeScript, deployed on AWS.
- Implemented authentication from the ground up, including Google OAuth login, across NestJS and FastAPI services.
- Set up the CI/CD pipeline for the NestJS backend using GitHub Actions — merges to main auto-deploy to AWS, removing manual release steps.
- Designed agentic workflows with LangGraph and RAG pipelines with webhook-based tool-calling, integrating Twilio telephony and PMS backends (OpenDental, Dentrix) for low-latency, context-aware voice interactions.

Jr. AI Developer | Erly Stage Studios 10/2024 – 06/2025 · Remote

- Built the full backend for the WhatsApp Meeting Scheduling Agent in FastAPI — webhook handling, agent orchestration (LangGraph + OpenAI), and state management.
- Deployed the service on GCP with Firestore as the primary data store; collaborated with an Anthropic engineer on system architecture and scalability decisions.
- Built a product Q&A chatbot API in Flask over scraped product data in PostgreSQL, combining direct SQL queries with pgvector similarity search for hybrid structured + semantic retrieval; handed off the API for frontend integration into the webapp.

AI & Data Analytics Intern | NCL NEDUET 04/2024 – 10/2024 · Karachi

- Delivered AI model development, EDA, and ML optimization projects with real-world business applications.
- Built LocoBot, a lab assistant robot combining real-time face recognition with a RAG chatbot for booking meetings with faculty; used directly by students visiting the lab.

KEY PROJECTS

Final Year Project — Novel Actor-Critic Algorithm for Reinforcement Learning

Reinforcement Learning · Actor-Critic · PyTorch · OpenAI Gym

Proposed a modification to the critic function in standard Actor-Critic architectures, achieving state-of-the-art performance surpassing prior baselines across multiple OpenAI Gym environments.

Enterprise Voice AI Agent

ElevenLabs · Twilio · LangChain · RAG · PMS

Intelligent appointment scheduler with RAG context retrieval, webhook tool-calling, and real-time telephony integration, designed for low-latency production use.

LocoBot — Face Recognition & Faculty Meeting Assistant

DeepFace · VGG-Face · Flask · SQLite · RAG

Lab robot that recognizes students in real time via a Flask API backend and uses a RAG chatbot to help them book meetings with faculty members — deployed for live use by students in the lab.

Violence Detection System

YOLOv8 · Computer Vision · Flask

Retrained YOLOv8 for real-time video analysis with streaming annotated frame alerts. Detects violent activity in the video feed, extracts the relevant clip, and can forward it to officials for review.

EDUCATION

B.E. Electrical Engineering — NED University of Engineering & Technology · CGPA: 2.5/4.0 · 02/2022 – 08/2026

Higher Secondary (Intermediate) — Bahria College Karsaz · BIEK Board · 996/1100 marks · 2020 – 2022

Secondary School (Matriculation) — CAA Model School and College, Karachi · BSEK Board · 755/850 marks · 2019 – 2020